

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

- a) What are the major differences between Microprocessor and Microcontroller? Why Harvard architecture is superior to Von-Neumann Architecture? 4+4
- b) What are the interrupt pins of 8085 microprocessor? Explain the different methods of handling multiple interrupts 7
- a) Write and assembly language program in 8085 to check whether the number stored in memory location D001 H is positive or negative. If positive store it in C001 else store in C002. Comments in program indicate full marks. 8
- b) What are the different types of machine cycles in 8085 instructions? Draw the labelled timing diagram for the instruction LDA 30DA H. 2+5
- a) Design an Addressing Decoding circuit to interface 2kB RAM, chips each of size 256 bytes with starting address 1001 H. 8
- b) Define interrupts. Differentiate between I/O mapped I/O and Memory mapped I/O 7
- a) Explain memory interfacing? How do you interface two chips of 4K RAM and one chips of 2K ROM of 8085 with and without fold back addresses, starting address is 8000H? 8
- b) Write an 8086 ALP to display the string "POKHARA UNIVERSITY" in reverse order. 7
- a) What are the different dedicated interrupt pointers in 8086 microprocessor? Draw and explain the block diagram of 8086 IVT table along with related memory address. 8
- b) What is Macro assembler? Differentiate between Macros and Procedure. 7

6. a) What are the significances of assembler directives in microprocessor? List all the assembler's directives of 8086 microprocessor and explain them with example. 4
- b) What is Synchronous and Asynchronous data transmission? Explain the RS232C standard for serial communication. 2+2
7. Write short notes on: (Any two) 2+2
 - a) SRAM and DRAM
 - b) RIM and SIM Instruction
 - c) RS 423A